microorganisms, toxins and extracellular viruses, neutralizing them or tagging them for destruction by phagocytes and other mechanisms.

The cellular (cell mediated) immunity consists of the T cells. Each T cell expresses antigen receptors called T cell receptors (TCRS). Unlike membrane bound antibody on B cells, which can recognize antigen alone, T cell receptors can recognize only antigen that is bound to MHC molecules. There are two major types of MHC molecules. Class I MHC molecules are expressed by all nucleated cells. Class II MHC molecules are expressed only by antigen presenting cells such as dendritic cells, macrophages and B cells. When a naive T cell encounters antigen combined with an MHC molecule on a cell, the T cell proliferates and differentiates into memory T cells and various effector T cells (helper T cells, cytotoxic T cells and regulatory T cells). Specific kinds of T cells directly attack target cells infected with viruses or parasites, transplanted cells or organs and cancer cells. T cells can induce target cell suicide (apoptosis), lyse targets cells, or release chemicals (cytokines) that enhance specific immunity and non specific defences such as phagocytosis and inflammation.

13.4.5 Types of Specific Immunity

Specific immunity can be acquired by natural means actively through infection or passively through receipt of preformed antibodies as through colostrum. Specific immunity can be acquired by artificial means actively through immunization or passively through receipt of preformed antibodies as with antisera.

13.5 Antigens

Substances capable of inducing a specific immune response are called antigens. The molecular properties of antigens and the way in which these properties ultimately contribute to immune activation are central to our understanding of the immune system.

13.5.1 Immunogenicity Versus Antigenicity

Two properties are exhibited by antigens; they are immunogenicity and antigenicity. Immunogenicity is the ability of an antigen to induce a humoral and / or cell mediated immune response.

B cells + antigen $\longrightarrow$ effector B cells (Plasma cells) + memory B cells

T cells + antigen $\longrightarrow$ effector T cells (T_C, T_H cells) + memory T cells

Although a substance that induces a specific immune response is usually called an antigen, it is more appropriately called an immunogen. Antigenicity is the ability of an antigen to combine specifically with the final products of the above responses. (antibodies and/or cell surface receptors). All immunogens are antigens but all antigens are not immunogens. Some small molecules called haptens are antigenic but incapable, by themselves, of inducing a specific immune response. In other words haptens lack immunogenicity. Examples of haptens are dinitrophenol, penicillin and m-amino benzene sulphonate.

13.5.2 Factors that Influence Immunogenicity

Immunogenicity is not an intrinsic property of an antigen but rather depends on a number of properties of the particular biological system that the antigen encounters. The

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